

# **Отчёт о лабораторной работе**

**Лабораторная работа 4**

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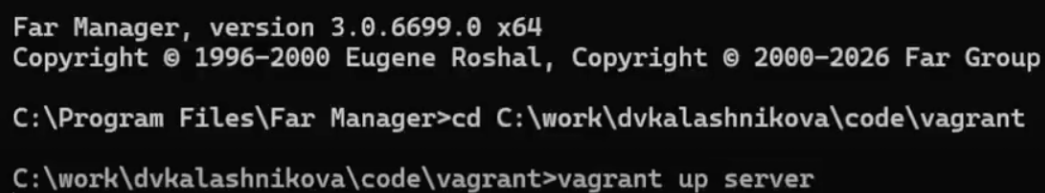
## **Список таблиц**

# 1 Цель работы

Приобретение практических навыков по установке и базовому конфигурированию HTTP-сервера Apache.

## 2 Выполнение лабораторной работы

Для начала запустим наш сервер через vagrant (рис. 2.1).



```
Far Manager, version 3.0.6699.0 x64  
Copyright © 1996-2000 Eugene Roshal, Copyright © 2000-2026 Far Group  
C:\Program Files\Far Manager>cd C:\work\dvkalashnikova\code\vagrant  
C:\work\dvkalashnikova\code\vagrant>vagrant up server
```

Рис. 2.1: Запуск сервера

Установим пакеты, необходимые для работы веб-сервера (рис. 2.2).

```
[dvkashnikova@server ~]$ sudo -i
[sudo] password for dvkashnikova:
[root@server dvkashnikova.net ~]# LANG=C yum grouplist
Extra Packages for Enterprise Linux 10 - x86_64
Extra Packages for Enterprise Linux 10 - x86_64
Rocky Linux 10 - BaseOS
Rocky Linux 10 - BaseOS
Rocky Linux 10 - AppStream
Rocky Linux 10 - AppStream
Rocky Linux 10 - CRB
Rocky Linux 10 - CRB
Rocky Linux 10 - Extras
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  KDE Plasma Workspaces
  KDE Plasma Mobile
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Desktop accessibility
  KDE Applications
  KDE
  KDE Multimedia support
  KDE Mobile
  KDE PIM
  KDE Software Development
  KDE Frameworks 6 Software Development
  Legacy UNIX Compatibility
  Smart Card Support
  Console Internet Tools
  .NET Development
  Graphical Administration Tools
  Network Servers
  RPM Development Tools
  Scientific Support
  Security Tools
  System Tools
[root@server dvkashnikova.net ~]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:00:57 ago on Thu 24 Sep 2026 03:48:33 PM UTC.
Dependencies resolved.

```

| Package                           | Architecture | Version              | Repository | Size  |
|-----------------------------------|--------------|----------------------|------------|-------|
| Installing group/module packages: |              |                      |            |       |
| httpd                             | x86_64       | 2.4.63-13.el10_2.6   | appstream  | 47 k  |
| httpd-manual                      | noarch       | 2.4.63-13.el10_2.6   | appstream  | 2.2 M |
| mod_fcgid                         | x86_64       | 2.3.9-35.el10        | appstream  | 72 k  |
| mod_ssl                           | x86_64       | 1:2.4.63-13.el10_2.6 | appstream  | 110 k |
| Installing dependencies:          |              |                      |            |       |
| apr                               | x86_64       | 1.7.5-3.el10         | appstream  | 127 k |
| apr-util                          | x86_64       | 1.6.3-23.el10_2.1    | appstream  | 98 k  |
| apr-util-ldap                     | x86_64       | 1.6.3-23.el10_2.1    | appstream  | 13 k  |
| httpd-core                        | x86_64       | 2.4.63-13.el10_2.6   | appstream  | 1.4 M |

Рис. 2.2: Установка пакетов

Проанализируем содержимое конфигурационных файлов сервера. Начнём с файла `/etc/httpd/conf/httpd.conf`. В нём содержатся основные настройки веб-сервера Apache (рис. 2.3).

```
GNU nano 8.1 /etc/httpd/conf/httpd.conf
# documents. By default, all requests are taken from this directory, but
# symbolic links and aliases may be used to point to other locations.
#
DocumentRoot "/var/www/html"

#
# Relax access to content within /var/www.
#
<Directory "/var/www">
    AllowOverride None
    # Allow open access:
    Require all granted
</Directory>

# Further relax access to the default document root:
<Directory "/var/www/html">
    #
    # Possible values for the Options directive are "None", "All",
    # or any combination of:
    #   Indexes Includes FollowSymLinks SymLinksifOwnerMatch ExecCGI MultiViews
    #
    # Note that "MultiViews" must be named *explicitly* --- "Options All"
    # doesn't give it to you.
    #
    # The Options directive is both complicated and important. Please see
    # http://httpd.apache.org/docs/2.4/mod/core.html#options
    # for more information.
    #
    Options Indexes FollowSymLinks

    #
    # AllowOverride controls what directives may be placed in .htaccess files.
    # It can be "All", "None", or any combination of the keywords:
    #   Options FileInfo AuthConfig Limit
    #
    AllowOverride None

    #
    # Controls who can get stuff from this server.
    #
    Require all granted
</Directory>

#
# DirectoryIndex: sets the file that Apache will serve if a directory
# is requested.
```

Рис. 2.3: /etc/httpd/conf/httpd.conf

В файле /etc/httpd/conf/magic содержатся инструкции для определения MIME-типа файла по его содержимому, а не расширению (рис. 2.4).

```

GNU nano 8.1 /etc/httpd/conf/magic
# Magic data for mod_mime_magic Apache module (originally for file(1) command)
# The module is described in /manual/mod/mod_mime_magic.html
#
# The format is 4-5 columns:
#   Column #1: byte number to begin checking from, ">" indicates continuation
#   Column #2: type of data to match
#   Column #3: contents of data to match
#   Column #4: MIME type of result
#   Column #5: MIME encoding of result (optional)
#-----
# Localstuff: file(1) magic for locally observed files
# Add any locally observed files here.
#-----
# end local stuff
#-----
#
# Java
0      short      0xcafe
>2     short      0xbabe      application/java
#-----
# audio: file(1) magic for sound formats
#
# from Jan Nicolai Langfeldt <janleifi.uio.no>,
#
# Sun/NeXT audio data
0      string      .snd
>12    belong      1          audio/basic
>12    belong      2          audio/basic
>12    belong      3          audio/basic
>12    belong      4          audio/basic
>12    belong      5          audio/basic
>12    belong      6          audio/basic
>12    belong      7          audio/basic
>12    belong      23         audio/x-adpcm
# DEC systems (e.g. DECstation 5000) use a variant of the Sun/NeXT format
# that uses little-endian encoding and has a different magic number
# (0x0064732E in little-endian encoding).
0      lelong      0x0064732E

```

Рис. 2.4: /etc/httpd/conf/magic

В файле /etc/httpd/conf.d/autoindex.conf содержатся настройки для автоматического отображения списка файлов в директории (рис. 2.5).

```
GNU nano 8.1 /etc/httpd/conf.d/autoindex.conf
#
# Directives controlling the display of server-generated directory listings.
#
# Required modules: mod_authz_core, mod_authz_host,
#                  mod_autoindex, mod_alias
#
# To see the listing of a directory, the Options directive for the
# directory must include "Indexes", and the directory must not contain
# a file matching those listed in the DirectoryIndex directive.
#
#
# IndexOptions: Controls the appearance of server-generated directory
# listings.
#
IndexOptions FancyIndexing HTMLTable VersionSort

# We include the /icons/ alias for FancyIndexed directory listings.  If
# you do not use FancyIndexing, you may comment this out.
#
Alias /icons/ "/usr/share/httpd/icons/"

<Directory "/usr/share/httpd/icons">
    Options Indexes MultiViews FollowSymlinks
    AllowOverride None
    Require all granted
</Directory>

#
# AddIcon* directives tell the server which icon to show for different
# files or filename extensions.  These are only displayed for
# FancyIndexed directories.
#
AddIconByEncoding (CMP,/icons/compressed.gif) x-compress x-gzip

AddIconByType (TXT,/icons/text.gif) text/*
AddIconByType (IMG,/icons/image2.gif) image/*
AddIconByType (SND,/icons/sound2.gif) audio/*
AddIconByType (VID,/icons/movie.gif) video/*
AddIconByType /icons/bomb.gif application/x-coredump

AddIcon /icons/binary.gif .bin .exe
AddIcon /icons/binhex.gif .hqx
AddIcon /icons/tar.gif .tar
AddIcon /icons/world2.gif .wrl .wrl.gz .vrm .iv
AddIcon /icons/compressed.gif .Z .z .tgz .gz .zip
```

Рис. 2.5: /etc/httpd/conf.d/autoindex.conf

В файле /etc/httpd/conf.d/manual.conf содержатся настройки для доступа к веб-странице с документацией Apache (рис. 2.6).

```
GNU nano 8.1 /etc/httpd/conf.d/manual.conf
#
# This configuration file allows the manual to be accessed at
# http://localhost/manual/
#
Alias /manual /usr/share/httpd/manual

<Directory "/usr/share/httpd/manual">
    Options Indexes
    AllowOverride None
    Require all granted

    RedirectMatch 301 ^/manual/(?:da|de|en|es|fr|ja|ko|pt-br|ru|tr|zh-cn)(/.*)$ "/manual$1"
</Directory>
```

Рис. 2.6: /etc/httpd/conf.d/manual.conf

В файле /etc/httpd/conf.d/userdir.conf содержатся настройки для доступа к публичным веб-директориям пользователей системы (рис. 2.7).

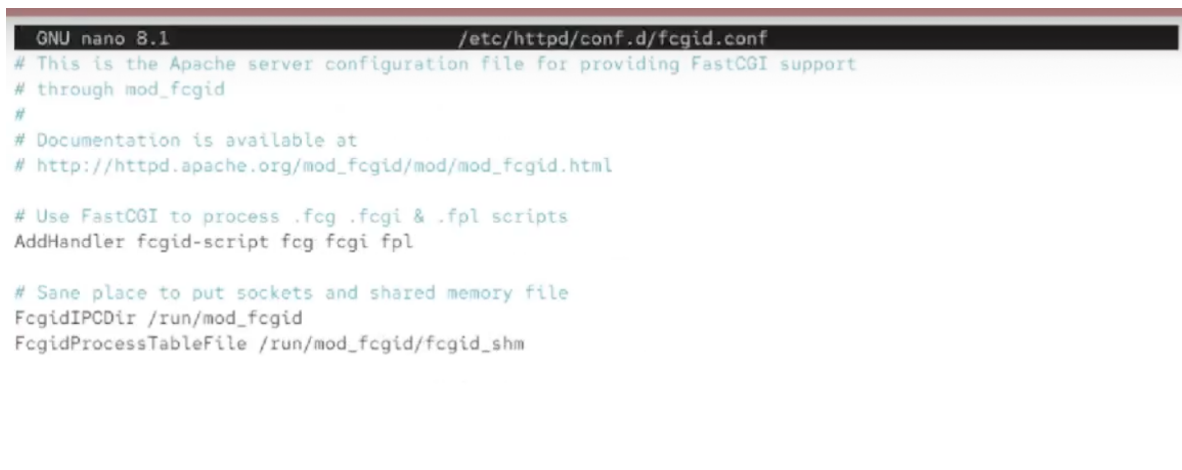
```
GNU nano 8.1 /etc/httpd/conf.d/userdir.conf
#
# UserDir: The name of the directory that is appended onto a user's home
# directory if a ~user request is received.
#
# The path to the end user account 'public_html' directory must be
# accessible to the webserver userid. This usually means that ~userid
# must have permissions of 711, ~userid/public_html must have permissions
# of 755, and documents contained therein must be world-readable.
# Otherwise, the client will only receive a "403 Forbidden" message.
#
<IfModule mod_userdir.c>
    #
    # UserDir is disabled by default since it can confirm the presence
    # of a username on the system (depending on home directory
    # permissions).
    #
    UserDir disabled

    #
    # To enable requests to ~/user/ to serve the user's public_html
    # directory, remove the "UserDir disabled" line above, and uncomment
    # the following line instead:
    #
    #UserDir public_html
</IfModule>

#
# Control access to UserDir directories. The following is an example
# for a site where these directories are restricted to read-only.
#
<Directory "/home/*/public_html">
    AllowOverride FileInfo AuthConfig Limit Indexes
    Options MultiViews Indexes SymLinksIfOwnerMatch IncludesNoExec
    Require method GET POST OPTIONS
</Directory>
```

Рис. 2.7: /etc/httpd/conf.d/userdir.conf

В файле /etc/httpd/conf.d/fcgid.conf содержатся настройки для модуля mod\_fcgid, который используется для запуска скриптов (например, PHP) через FastCGI (рис. 2.8).



```
GNU nano 8.1 /etc/httpd/conf.d/fcgid.conf
# This is the Apache server configuration file for providing FastCGI support
# through mod_fcgid
#
# Documentation is available at
# http://httpd.apache.org/mod_fcgid/mod/mod_fcgid.html

# Use FastCGI to process .fcg .fcgi & .fpl scripts
AddHandler fcgid-script fcg fcgi fpl

# Sane place to put sockets and shared memory file
FcgidIPCDir /run/mod_fcgid
FcgidProcessTableFile /run/mod_fcgid/fcgid_shm
```

Рис. 2.8: /etc/httpd/conf.d/fcgid.conf

В файле /etc/httpd/conf.d/ssl.conf содержатся настройки для поддержки шифрования SSL/TLS (рис. 2.9).

```
GNU nano 8.1 /etc/httpd/conf.d/ssl.conf
##
# When we also provide SSL we have to listen to the
# standard HTTPS port in addition.
#
Listen 443 https

##
##  SSL Global Context
##
##  All SSL configuration in this context applies both to
##  the main server and all SSL-enabled virtual hosts.
##

#   Pass Phrase Dialog:
#   Configure the pass phrase gathering process.
#   The filtering dialog program ('builtin' is a internal
#   terminal dialog) has to provide the pass phrase on stdout.
SSLPassPhraseDialog exec:/usr/libexec/httpd-ssl-pass-dialog

#   Inter-Process Session Cache:
#   Configure the SSL Session Cache: First the mechanism
#   to use and second the expiring timeout (in seconds).
SSLSessionCache         shmcb:/run/httpd/sslcache(512000)
SSLSessionCacheTimeout  300

#
# Use "SSLCryptoDevice" to enable any supported hardware
# accelerators. Use "openssl engine -v" to list supported
# engine names. NOTE: If you enable an accelerator and the
# server does not start, consult the error logs and ensure
# your accelerator is functioning properly.
#
SSLCryptoDevice builtin
#SSLCryptoDevice ubsec

##
##  SSL Virtual Host Context
##

<VirtualHost _default_:443>

# General setup for the virtual host, inherited from global configuration
#DocumentRoot "/var/www/html"
#ServerName www.example.com:443
```

Рис. 2.9: /etc/httpd/conf.d/ssl.conf

Теперь добавим службу http в фаервол для корректной работы (рис. 2.10).

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
bash: firewall-cmd: command not found...
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns ssh
[root@server.dvkalashnikova.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-clien
t ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 ip
erf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogi
n kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controll
er-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kub
e-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr l
lmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mongodb mosh mou
ntd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-
dashboard nfs nfs3 nmea-0183 nrip ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-
vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius radsec rdp red
is redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane set
tlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptr
ap socks spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming stellar
is stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-rel
ay synergy syscomlan syslog-ng syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks tran
smission-client turn turns upnp-client vdsd vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-
discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman ws
mans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server za
bbix-trapper zabbix-web-service zero-k zerotier
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http --permanent
bash: firewall-cmd: command not found...
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http --permanent
success
```

Рис. 2.10: Настройка firewall

Запустим в отдельной вкладке `journalctl` с ключом `-f` для вывода информации логов в реальном времени (рис. 2.11).

```
Subject: Process 18242 (VBoxClient) dumped core
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support
Documentation: man:core(5)

Process 18242 (VBoxClient) crashed and dumped core.

This usually indicates a programming error in the crashing program and
should be reported to its vendor as a bug.
Sep 24 15:57:59 server.dvkalashnikova.net systemd[1]: systemd-coredump@202-18246-0.service: Deactivating successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-coredump@202-18246-0.service has successfully entered the 'dead' state.
```

Рис. 2.11: journalctl

Теперь попробуем запустить службу httpd и включить в ней автозагрузку (рис. 2.12).

```
[root@server.dvkalashnikova.net ~]# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
[root@server.dvkalashnikova.net ~]# systemctl start httpd
```

Рис. 2.12: запуск службы httpd

Вернёмся ко вкладке с journalctl. Как видим, запуск httpd был успешен (рис. 2.13).

```

should be reported to its vendor as a bug.
Sep 24 15:58:38 server.dvkalashnikova.net systemd[1]: systemd-coredump@208-18711-0.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-coredump@208-18711-0.service has successfully entered the 'dead' state.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit serial-getty@ttyS0.service has successfully entered the 'dead' state.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 142.
Subject: Automatic restarting of a unit has been scheduled
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

Automatic restarting of the unit serial-getty@ttyS0.service has been scheduled, as the result for the configured Restart= setting for the unit.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Subject: A start job for unit serial-getty@ttyS0.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit serial-getty@ttyS0.service has finished successfully.

The job identifier is 32606.

```

Рис. 2.13: Лог об успешном запуске

Теперь запустим клиент (рис. 2.14).

```

server: Retabaled /etc/networkmanager/system-connections
C:\work\dvkalashnikova\code\vagrant> vagrant up client

```

Рис. 2.14: Запуск клиента

Зайдём в клиенте в браузер и перейдём по адресу 192.168.1.1. Как видим, это страница, которая используется вебсервером по умолчанию. Таким образом, она даёт нам понять, что Служба httpd работает корректно, и клиент может получить доступ к содержимому веб-страницы (рис. 2.15).

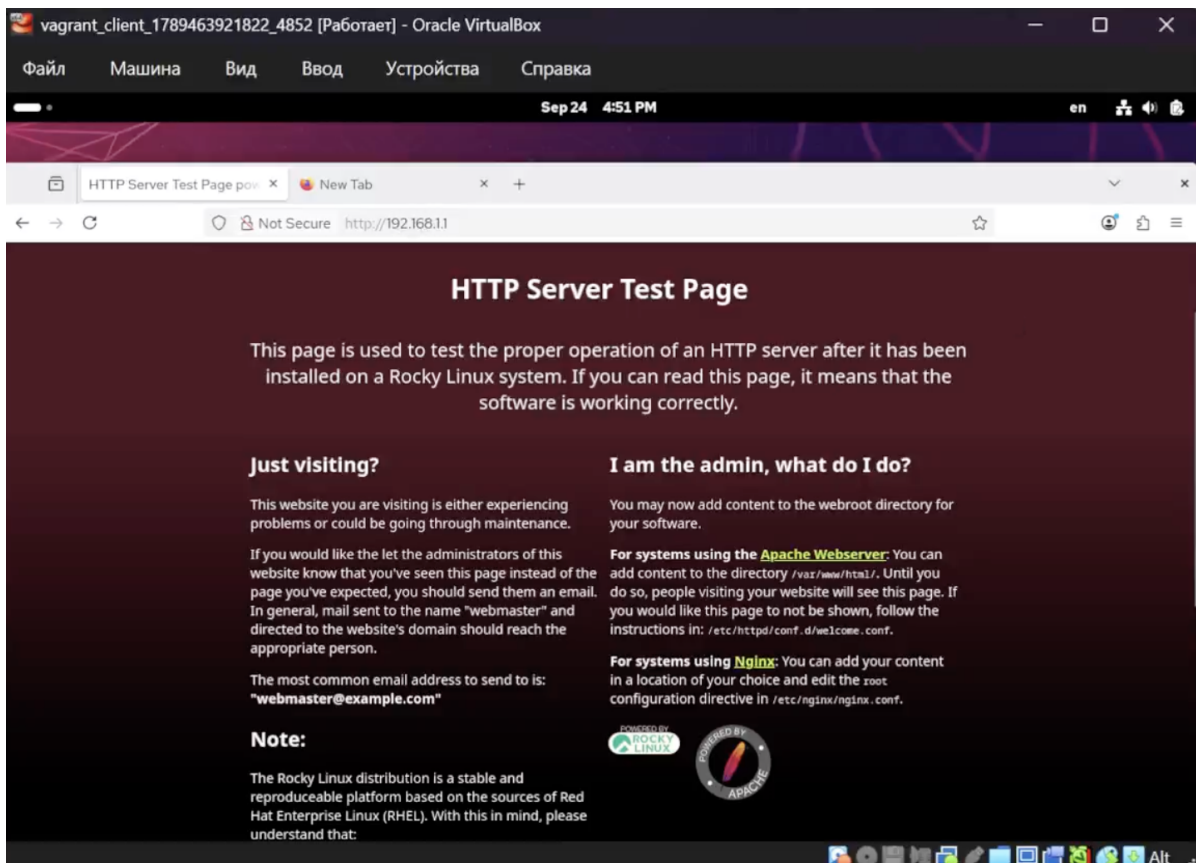


Рис. 2.15: Страница по умолчанию

Теперь выведем логи об ошибках веб-сервера в файле `/var/log/httpd/error_log` (рис. 2.16).

```
[dvkashnikova@server.dvkashnikova.net ~]$ sudo -i
[sudo] password for dvkashnikova:
[root@server.dvkashnikova.net ~]# tail -f /var/log/httpd/error_log
[Thu Sep 24 15:58:35.743830 2026] [suexec:notice] [pid 18525:tid 18525] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)
[Thu Sep 24 15:58:35.976442 2026] [lbmethod_heartbeat:notice] [pid 18525:tid 18525] AH02282: No slotmem from mod_heartbeat
[Thu Sep 24 15:58:36.001799 2026] [systemd:notice] [pid 18525:tid 18525] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0
[Thu Sep 24 15:58:36.046646 2026] [mpm_event:notice] [pid 18525:tid 18525] AH00489: Apache/2.4.63 (Rocky Linux) OpenSSL/3.5.5 mod_fcgid/2.3.9 configured -- resuming normal operations
[Thu Sep 24 15:58:36.046721 2026] [core:notice] [pid 18525:tid 18525] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
```

Рис. 2.16: `/var/log/httpd/error_log`

И выведем лог о доступе к веб-странице в файле `/var/log/httpd/access_log` (рис.

2.17).

```
[dvkalashnikova@server.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkalashnikova:
[root@server.dvkalashnikova.net ~]# tail -f /var/log/httpd/access_log
192.168.1.2 - - [24/Sep/2026:16:48:42 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0 (X
11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.2 - - [24/Sep/2026:16:48:43 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15443
"http://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/1
40.0"
192.168.1.2 - - [24/Sep/2026:16:48:43 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "http:
//192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.2 - - [24/Sep/2026:16:48:44 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://1
92.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
```

Рис. 2.17: /var/log/httpd/access\_log

Остановим службу named, отвечающую за dns (рис. 2.18).

```
[root@server.dvkalashnikova.net ~]# systemctl stop named
[root@server.dvkalashnikova.net ~]#
```

Рис. 2.18: Остановка службы named

Поменяем файл зоны, добавив в него запись о том, что теперь у нас есть http сервер (www) (рис. 2.19).

```
GNU nano 8.1 /var/named/master/fz/dvkalashnikova.net Modified
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                2026091501      ; serial
                                1D                ; refresh
                                1H                ; retry
                                1W                ; expire
                                3H )              ; minimum

      NS      @
      A       192.168.1.1
      AAAA    ::1
$ORIGIN dvkalashnikova.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
www A 192.168.1.1
```

Рис. 2.19: Файл зоны /var/named/master/fz/dvkalashnikova

То же самое сделаем в обратном файле зоны (рис. 2.20).

```
GNU nano 8.1 /var/named/master/rz/192.168.1 Mo
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                2026091501      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

      NS      @
      A       192.168.1.1
      AAAA    ::1
      PTR     server.dvkalashnikova.net
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.dvkalashnikova.net.
1 PTR ns.dvkalashnikova.net.
1 PTR dhcp.dvkalashnikova.net.
1 PTR www.dvkalashnikova.net.
```

Рис. 2.20: Файл зоны /var/named/master/rz/192.168.1

Удалим файлы журналов из папок обратной и прямой зон (рис. 2.21).

```
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/192.168.1
[root@server.dvkalashnikova.net ~]# nano /var/named/master/rz/192.168.1
[root@server.dvkalashnikova.net ~]# ls /var/named/master/fz/
dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# ls /var/named/master/fz/
dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# ls /var/named/master/rz/
192.168.1
```

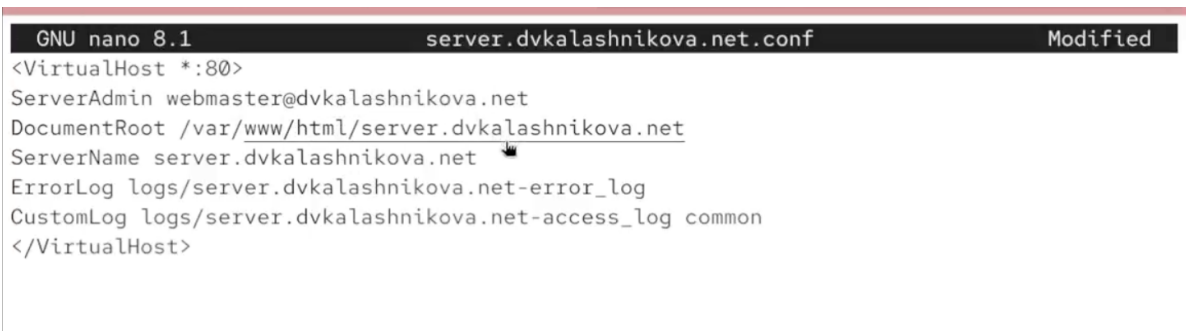
Рис. 2.21: Удаление журналов

Теперь запустим dns службу и перейдём в папку с конфигурацией httpd (/etc/httpd/conf.d), создав там файлы конфигурации для двух страниц - server.dvkalashnikova.net и www.dvkalashnikova.net (рис. 2.22).

```
[root@server.dvkalashnikova.net ~]# systemctl start named
[r File display r.dvkalashnikova.net ~]# cd /etc/httpd/conf.d
[root@server.dvkalashnikova.net conf.d]# ouch server.dvkalashnikova.net.conf
bash: ouch: command not found...
[root@server.dvkalashnikova.net conf.d]# touch server.dvkalashnikova.net.conf
[root@server.dvkalashnikova.net conf.d]# touch www.dvkalashnikova.net.conf
[root@server.dvkalashnikova.net conf.d]# nano server.dvkalashnikova.net.conf
```

Рис. 2.22: Создание конфигурационных файлов

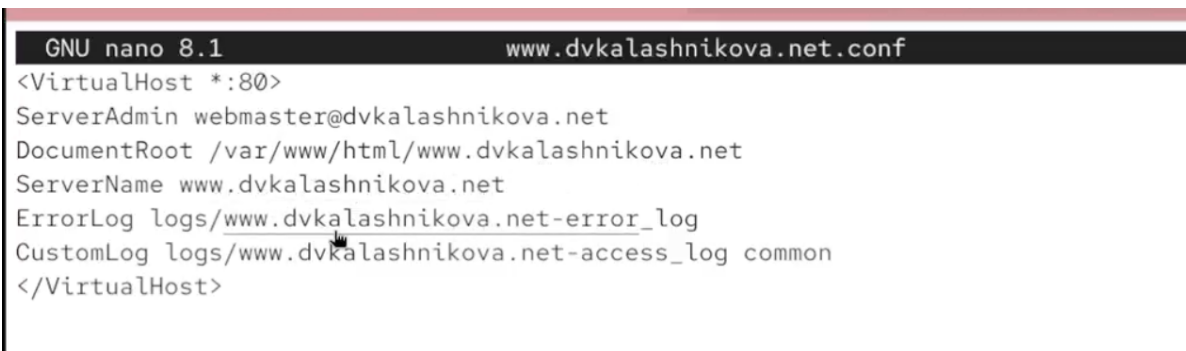
Поместим следующее содержимое в первый из файлов (рис. 2.23).



```
GNU nano 8.1 server.dvkalashnikova.net.conf Modified
<VirtualHost *:80>
ServerAdmin webmaster@dvkalashnikova.net
DocumentRoot /var/www/html/server.dvkalashnikova.net
ServerName server.dvkalashnikova.net
ErrorLog logs/server.dvkalashnikova.net-error_log
CustomLog logs/server.dvkalashnikova.net-access_log common
</VirtualHost>
```

Рис. 2.23: server.dvkalashnikova.net

И следующее содержимое во второй файл (рис. 2.24).



```
GNU nano 8.1 www.dvkalashnikova.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@dvkalashnikova.net
DocumentRoot /var/www/html/www.dvkalashnikova.net
ServerName www.dvkalashnikova.net
ErrorLog logs/www.dvkalashnikova.net-error_log
CustomLog logs/www.dvkalashnikova.net-access_log common
</VirtualHost>
```

Рис. 2.24: www.dvkalashnikova.net

В папке /var/www/html создадим папки server.dvkalashnikova.net и www.dvkalashnikova.net. В каждой из них создадим файл index.html, в каждый из которых запишем простую приветственную фразу (рис. 2.25).

```
[root@server.dvkalashnikova.net conf.d]# cd /var/www/html
[root@server.dvkalashnikova.net html]# mkdir server.dvkalashnikova.net
[root@server.dvkalashnikova.net html]# cd /var/www/html/server.dvkalashnikova.net
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# touch index.html
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# echo "Welcome to the server.dvkalashnikova.net server." > index.html
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# cd /var/www/html
[root@server.dvkalashnikova.net html]# mkdir www.dvkalashnikova.net
[root@server.dvkalashnikova.net html]# cd /var/www/html/www.dvkalashnikova.net
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# touch index.html
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# echo "Welcome to the www.dvkalashnikova.net server." > index.html
```

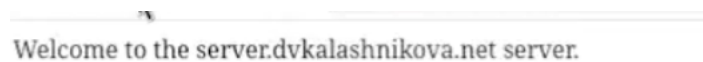
Рис. 2.25: Файлы index.html

Теперь обновим метки для SELinux и перезапустим службу httpd (рис. 2.26).

```
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# chown -R apache:apache /var/www
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /etc
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /var/named
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /var/www
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# systemctl restart httpd
```

Рис. 2.26: Обновление меток и перезапуск службы

Теперь с клиента попробуем перейти по адресу server.dvkalashnikova.net. Как видим, нам вывело страницу с той самой фразой, которую мы записывали в index.html ранее (рис. 2.27).



Welcome to the server.dvkalashnikova.net server.

Рис. 2.27: server.dvkalashnikova.net

Теперь обновим конфигурацию vagrant, поместив обновлённые конфиги и все созданные в ходе лабораторной работы файлы в папку /vagrant/provision/server. Кроме того, создадим скрипт http.sh (рис. 2.28).

```
[root@server.dvkalashnikova.net server]# cd /vagrant/provision/server/dns/
[root@server.dvkalashnikova.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/
named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/dvkalashnikova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol
error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Pro
tocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost':
Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback':
Protocol error
[root@server.dvkalashnikova.net dns]# cd /vagrant/provision/server
[root@server.dvkalashnikova.net server]# touch http.sh
[root@server.dvkalashnikova.net server]# chmod +x http.sh
```

Рис. 2.28: Обновление конфигурации vagrant

В скрипт `http.sh` пропишем следующее содержимое, позволяющее настроить наш http сервер (рис. 2.29).

```
GNU nano 8.1                                     http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

Рис. 2.29: Скрипт http.sh

Отредактируем Vagrantfile, добавив в его конфигурацию запуск созданного ранее скрипта (рис. 2.30).

```
server.vm.provision "server dummy", type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"
server.vm.provision "server forward", type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forward.sh"
server.vm.provision "server dns", type: "shell", preserve_order: true, path: "provision/server/dns.sh"
server.vm.provision "server dhcp", type: "shell", preserve_order: true, path: "provision/server/dhcp.sh"
server.vm.provision "server http", type: "shell", preserve_order: true, path: "provision/server/http.sh"
end
```

Рис. 2.30: Vagrantfile

## **3 Выводы**

В результате выполнения лабораторной работы были получены навыки работы и настройки http сервера